Mushroom Classification – Variable Cheat-Sheet

Hand-off note for the team. Use these variables to build more models without re-reading the code.

1. The key thing for modeling

Models take the fully preprocessed matrix (scaled + one-hot, no NaN) plus the target:

```
model.fit(X_train_prep, y_train)
model.predict(X_test_prep)
model.score(X_test_prep, y_test)
```

Variable	What it holds	Old name
`X_train_prep / X_test_prep`	model-ready: StandardScaler numbers + one-hot categories, no NaN	old X_train / X_test (but get_dummies + unscaled)
`X_train_robprep / X_test_robprep`	same, but with RobustScaler (only if you want to test robust)	(new)
`y_train / y_test`	target: 0 = edible, 1 = poisonous	same (source was ys)

Default = X_train_prep (Standard ~= Robust; Standard marginally ahead).

2. Gotcha – X_train now means something different!

Variable	What it holds NOW	Old name
`X / y`	raw features / target from the original df	used to be Xs / ys
`X_train / X_test`	RAW: categories as text + NaN (we split BEFORE encoding)	same name, new content

Old code that fed X_train straight into a model now breaks (strings + NaN). Replace X_train -> X_train_prep, X_test -> X_test_prep.

3. For EDA (readable, training data only)

Variable	What it holds	Old name
`df_train`	training rows: readable categories + class + NaN -> patterns / plots	(new)
`df_metr_nan`	numbers 0/1/2 only, with NaN	df_metrical

Do EDA only on df_train. Never on df (contains test) or X_test (the vault).

4. Intermediate steps (if you build your own preprocessing)

Variable	What it holds
`num_colum / categ_colum`	column lists: ['0','1','2'] / the 5 categorical columns
`X_train_median / X_test_median`	numbers after median-impute (still unscaled)
`X_train_scaled / X_train_robscaled`	numbers impute + Standard / Robust scaler
`X_train_onehot / X_test_onehot`	categories (imputed 'MISSING') one-hot encoded

How the model matrix is built: X_train_prep = np.hstack([X_train_scaled, X_train_onehot])

5. Fitted transformers (to treat NEW data the same way)

scaler (StandardScaler), robscaler (RobustScaler), onehot (OneHotEncoder).

Note: imputer is used twice (median for numbers, then 'MISSING' for categories) - it ends up pointing at the category imputer.

6. Models already built

logreg (LogReg), knn (kNN + StandardScaler), knn2 (kNN + RobustScaler).